A MAGIS-100 Experimental Forecast for Momentum-Quadratic Dephasing

A Short Technical Note on Experimental Discriminants for a Parameter-Free Transport Prediction

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Abstract

This note translates the momentum-quadratic dephasing prediction of null-directional kinematics into an experimentally oriented forecast for MAGIS-100. The goal is not to rederive the underlying theory, but to express the predicted signature in the variables most relevant to a long-baseline clock atom interferometer and to identify concrete control tests that distinguish it from standard technical loss channels.

The central prediction is a residual dephasing component whose leading dependence is quadratic in large-momentum-transfer order, linear in interrogation time, and stronger for smaller pre-split coherence length or, under the thermal identification, for higher source temperature. Unlike earlier formulations, the overall normalization of this effect is not set by an external ansatz but is fixed internally by the primitive-cell sector of the theory through the action-scale closure.

The sharpest content of the forecast is therefore the joint scaling law across multiple independent experimental control variables together with a normalization fixed by the primitive-cell sector rather than by an external ansatz. The note proposes a compact discrimination program built from LMT sweeps, source-preparation sweeps, interrogation-time sweeps, and ordinary pulse recalibration. Its main experimental value is not merely a positive forecast, but a falsifiable strategy for separating a substrate-driven coherence burden from ordinary technical contrast loss.

1 Purpose and scope

MAGIS-100 is designed to operate as a long-baseline clock atom interferometer in a regime where large momentum transfer (LMT), long free-fall times, and large arm separations are all central design features [1]. That makes it a natural target for testing whether contrast degradation contains a contribution beyond ordinary technical pulse inefficiency or classical noise.

The present note is therefore intentionally narrow. It does not attempt to restate the full NDK framework or the derivation of emergent Schrödinger dynamics. Those tasks are handled in the upstream papers [2, 3]. Instead, the goal here is to provide an experimentally oriented forecast based on the LMT dephasing Letter [4]:

(i) express the predicted signature in observable interferometer variables;

(ii) state how it differs from standard LMT loss channels;

(iii) identify simple sweeps that could support or disfavor the effect.

The sharpest content of the present forecast is the functional form of the residual scaling law rather than the absolute value of its overall coefficient. The note predicts a joint dependence on LMT order, interrogation time, and source preparation that is not shared by standard technical loss channels. In the present formulation, the overall normalization is fixed internally by the primitive-cell sector of the theory through the action-scale closure, so that the prediction contains no free normalization parameter.

MAGIS-100 remains under active construction and development as a long-baseline clock atom interferometer and pathfinder instrument [1, 5].

2 Prediction in experimental variables

2.1 Residual dephasing law in LMT-letter notation

After standard technical-noise and environmental modeling, define the residual visibility by

$$V_{\rm res} \equiv \frac{\mathcal{C}_{\rm obs}}{\mathcal{C}_{\rm tech}}, \tag{1}$$

and the associated residual dephasing exponent by

$$D_{\rm pred} \equiv -\ln V_{\rm res}. \tag{2}$$

In the transport formulation of [4], the cell-level transport burden is

$$\mathcal{D}_{\rm tr} \equiv \pi\,\chi\,\mathcal{I}_*/A\,c\,\ell_{\rm P}\,T\,n^2 k_{\rm eff}^2, \tag{3}$$

where T is the interrogation time, n is the effective LMT order, $k_{\rm eff}$ is the effective wavevector, χ is an order-unity geometric factor, and $\mathcal{I}_*/A$ is the adopted directional-strain benchmark.

The operational coherence-cell area is identified with the primitive cell area of the directional substrate. The action-scale closure establishes that the emergent quantum of action satisfies

$$\hbar_{\rm eff} = \frac{c^3}{G} A_*, \tag{4}$$

so that the normalization of the transport-induced dephasing is fixed by the same primitive-cell sector that defines the underlying phase structure [6]. In particular, the closure identifies the primitive support cell with a Planck-area normalization, replacing the earlier fixed-cell ansatz by a closure-normalized geometric scale.

Accordingly, the predicted residual dephasing does not depend on an external ansatz for the cell size. What remains is the laboratory transfer map between coherence-preparation parameters and the effective experimental realization of that closure-normalized scale.

The most robust content of the forecast is both the shape of the residual scaling law and its normalization. The scaling structure follows from the transport law, while the overall coefficient is fixed by the primitive-cell sector through the action-scale closure. Remaining factors such as χ, $\xi_{\rm th}$, and laboratory coherence-transfer quantities enter as geometric or preparation-dependent transfer parameters, not as free normalization constants.

2.2 Temperature scaling under a thermal coherence identification

As a working thermal identification, take

$$\lambda_{\mathrm{coh},0}^2 = \xi_{\mathrm{th}}^2\,\lambda_{\mathrm{th}}^2, \qquad \lambda_{\mathrm{th}}^2 = \frac{2\pi\hbar^2}{m\,k_B\Theta}, \tag{5}$$

where Θ is the source temperature, m is the atomic mass, and ξ_{th} is an order-unity bridge factor.

Substituting equation (5) into the closure-normalized dephasing law gives

$$D_{\mathrm{pred}}(\Theta) = \frac{\chi}{2\eta\,\xi_{\mathrm{th}}^2}\,\mathcal{I}_*/A\,c\,\ell_{\mathrm{P}}\,T\,n^2 k_{\mathrm{eff}}^2\,w_{\mathrm{coh}}^2\,\frac{m\,k_B\Theta}{\hbar^2}. \tag{6}$$

Thus, in the fit-law language

$$D_{\mathrm{pred}}(n) = a_0 + a_1 n^2, \tag{7}$$

the quadratic dephasing coefficient satisfies

$$a_1(T,\Theta) \propto T\,\Theta \tag{8}$$

at fixed geometry and fixed transverse coherence-support width.

In the forecast figure below, this manifests as a vertical rescaling of the underlying n-dependence. At fixed geometry and controlled transverse coherence-support width, varying the source temperature Θ changes the overall coefficient of the residual law without altering its log-log shape as a function of n.

The resulting closure-normalized forecast is shown in figure 1, which makes clear that the primary experimental opportunity lies not in an overwhelmingly large signal at nominal settings, but in the onset of a structured residual near the sensitivity window of MAGIS-100.

2.3 What the prediction means operationally

The forecast is therefore not simply "contrast gets worse at larger LMT." Many standard mechanisms do that. The claim is more specific:

(a) the additional loss enters as a dephasing exponent rather than merely a cumulative pulse-population failure;

(b) it scales quadratically with the LMT enhancement variable;

(c) it strengthens with longer interrogation time;

(d) it strengthens when the pre-split coherence length $\lambda_{\mathrm{coh},0}$ is reduced, equivalently when $(w_{\mathrm{coh}}/\lambda_{\mathrm{coh},0})^2$ increases.

Those four features, taken together, are the real experimental signature.

In the plotted forecast, the remaining shaded spread should be read as a model-side sensitivity band generated by the residual sweep over $(w_{\mathrm{coh}}, \lambda_{\mathrm{coh},0})$, not as a statistical error bar or an experimental uncertainty estimate.

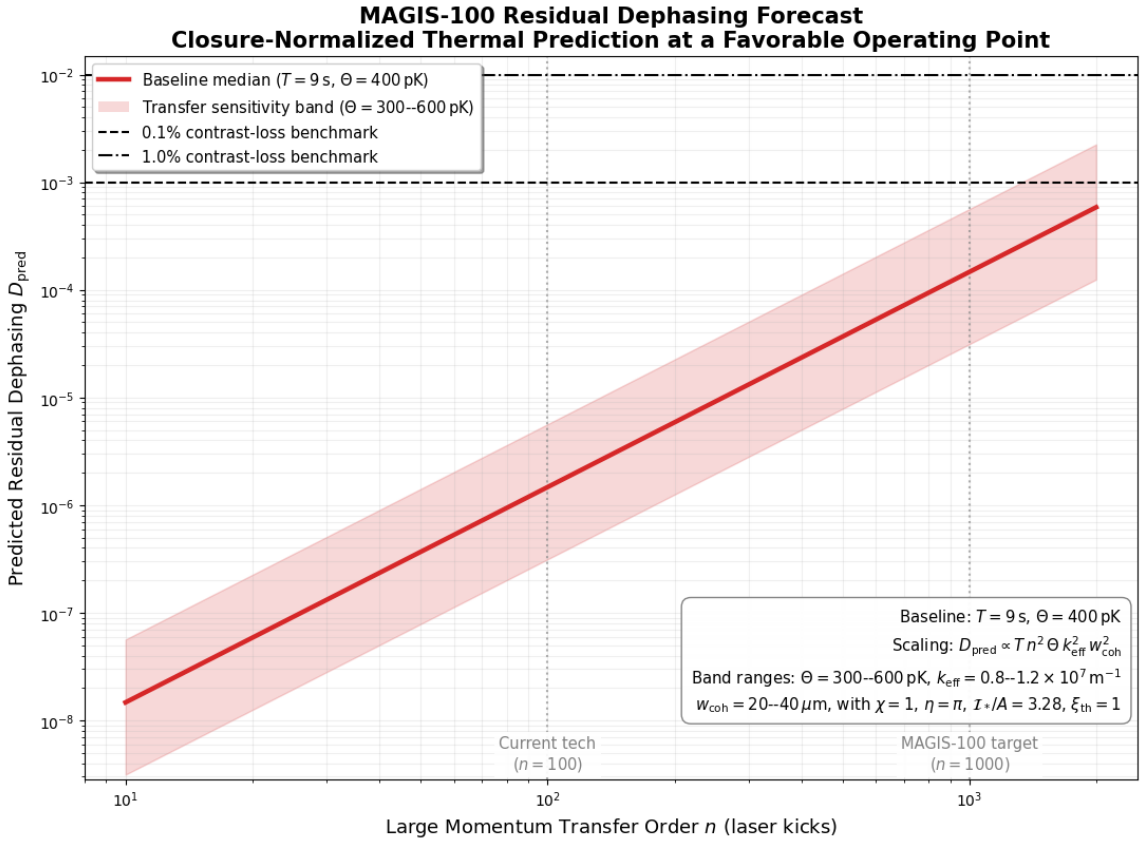

Figure 1: **MAGIS-100 closure-normalized residual dephasing forecast.** The forecast is shown as a function of large-momentum-transfer order n using the closure-normalized thermal scaling law. The baseline curve is evaluated at a favorable operating point with $T = 9\,\mathrm{s}$ and $\Theta = 400\,\mathrm{pK}$, while the shaded band indicates the corresponding sensitivity to laboratory transfer parameters. The horizontal black lines mark approximate 0.1% and 1% contrast-loss benchmarks. Under conservative settings the predicted residual remains below the sub-percent window, but under favorable preparation it approaches practical visibility near the $n \sim 10^3$ regime relevant to MAGIS-100.

3 Why this is not just an ordinary technical loss channel

Recent work on LMT clock interferometry has emphasized that several apparent scaling concerns are milder than once feared. In particular, sequential state inversions from alternating directions lead to cumulative population error that scales linearly with pulse number rather than quadratically, and laser-frequency noise need not be a practical blocker for high-fidelity LMT clock interferometers [7]. Likewise, broader LMT studies continue to frame imperfect pulse efficiencies, parasitic paths, and operating-regime optimization as central technical limits [8].

That is precisely why the present prediction is experimentally interesting: it is not a claim about ordinary pulse fidelity accumulation. The proposed dephasing burden arises from the transported phase-gradient load borne by the split matter-wave branches in the reduced sector, together with the primitive-cell normalization that converts that burden into an effective contrast floor [4, 3]. It should therefore be tested against, not conflated with, standard pulse-imperfection channels.

A compact way to summarize the distinction is:

$$D_{\mathrm{tech}} = D_{\mathrm{pulse}} + D_{\mathrm{wavefront}} + D_{\mathrm{inhom}} + D_{\mathrm{readout}} + \cdots , \tag{9}$$

whereas the closure-normalized transport contribution predicts

$$D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2 . \tag{10}$$

The useful experimental question is therefore not whether total contrast loss exists, but whether a residual component with the scaling of equation (10) survives after technical channels are characterized.

4 A minimal discrimination program for MAGIS-100

4.1 LMT sweep

At fixed source preparation and fixed interrogation time, measure the contrast as a function of the LMT control parameter. The key question is whether the residual dephasing exponent extracted from

$$-\ln\left(\frac{\mathcal{C}_{\mathrm{obs}}}{\mathcal{C}_{\mathrm{tech}}}\right)$$

is better fit by a linear law or a quadratic law in n after standard technical models are accounted for.

A clean quadratic residual would be the first nontrivial support for the predicted transport signature.

4.2 Source-preparation sweep

At fixed n and fixed T, vary the source preparation in a way that changes the pre-split coherence length $\lambda_{\mathrm{coh},0}$, or, under the thermal identification, the source temperature Θ. The model predicts:

$$D_{\mathrm{pred}} \propto \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2 \qquad \text{or, under the thermal identification,} \qquad D_{\mathrm{pred}} \propto \Theta. \tag{11}$$

This is one of the most important discriminants, because many ordinary pulse-calibration channels are not expected to track the pre-split coherence length $\lambda_{\mathrm{coh},0}$ or source temperature Θ in this direct way.

The temperature sweep is especially valuable because it is already part of the present derived phenomenological bridge, not an external speculative discriminator. By contrast, a possible future orientation-dependent refinement of the experimental geometry factor χ may be worth testing as an auxiliary cross-check, but the present note does not require or derive a quantitative angular law for that effect.

The most practical preparation-side lever within the present MAGIS-100 architecture is illustrated in figure 2, which shows how the predicted residual increases as the transverse coherence-support width and source temperature are increased within a favorable ultracold operating regime.

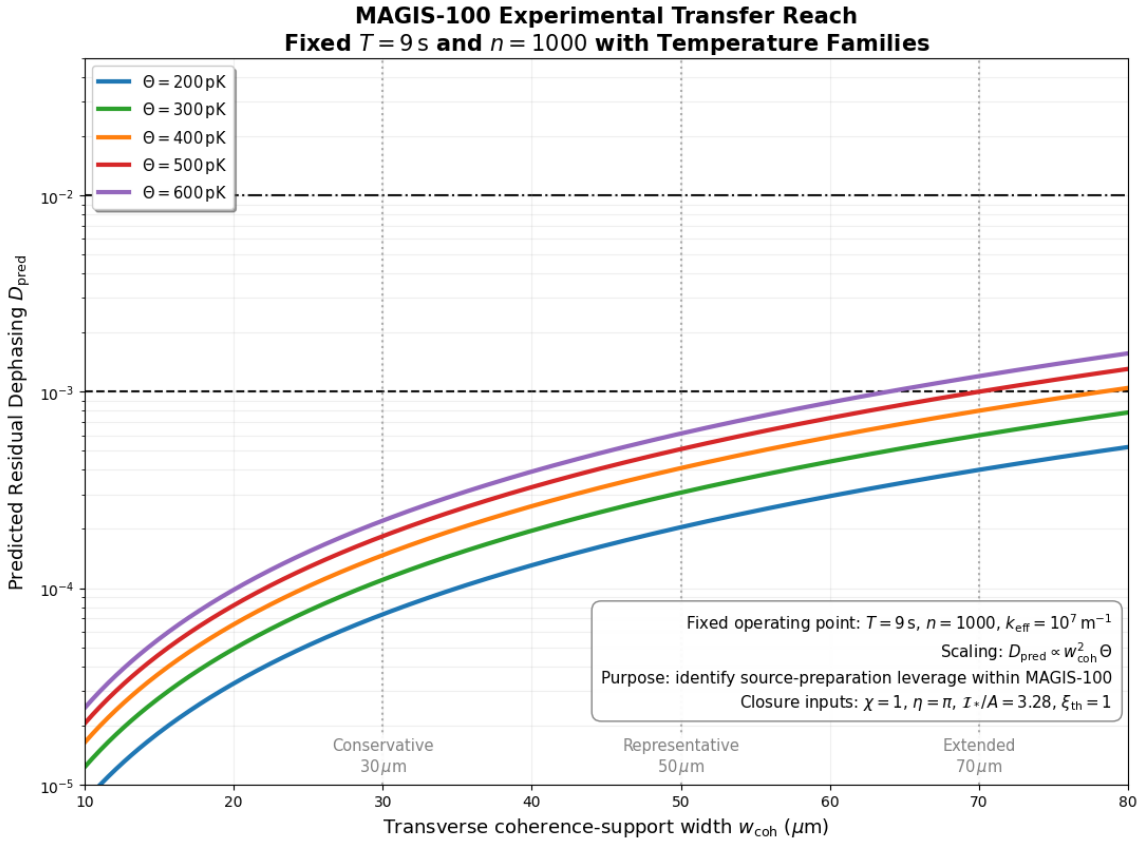

Figure 2: **Experimental transfer reach within the MAGIS-100 architecture.** This figure fixes the operating point at large interrogation time and high LMT order and shows the predicted residual dephasing as a function of transverse coherence-support width $w_{\rm coh}$ for a family of source temperatures. The labeled reference widths identify conservative, representative, and extended preparation regimes rather than fixed instrument design values. Because the closure-normalized prediction scales quadratically with $w_{\rm coh}$ and linearly with source temperature, modest relaxation of source-preparation parameters materially improves experimental reach.

The action-scale closure implies that the NDK vacuum anomaly is strongly suppressed under standard interferometric operating conditions, naturally accounting for the absence of any obvious signal in present 10-meter-scale atom interferometers. Even within the 100-meter baseline of MAGIS-100, the effect is not expected to appear as a dominant background under conservative ultracold operation. Instead, the present framework identifies MAGIS-100 as the first facility capable of carrying out a targeted search in a regime where the signal approaches practical visibility. In particular, the predicted residual is enhanced by broader transverse coherence-support width and warmer source preparation, providing a concrete experimental pathway for probing the closure-normalized transport signature above the ordinary technical floor.

4.3 Interrogation-time sweep

At fixed preparation and fixed n, the fixed-cell model predicts an approximately linear growth with interrogation time,

$$D_{\rm pred} \propto T. \tag{12}$$

A residual contribution that grows with T while keeping its preparation dependence would support the interpretation of the effect as a genuine branch-burden dephasing process rather than a one-shot pulse artifact.

4.4 Recalibration and pulse-optimization sweep

Technical pulse-loss mechanisms should be sensitive to:

- improved pulse shaping,
- reduced laser-frequency noise,
- better control of parasitic paths,
- operating-point optimization.

If a contrast-loss component remains after those improvements and continues to follow the scaling of equation (10), that residual becomes especially interesting.

Conversely, if the apparent n^2 trend disappears under ordinary pulse optimization, the fixed-cell interpretation is disfavored.

5 A practical discrimination summary

To isolate the proposed signature in the laboratory, the defining behaviors of the geometric coherence burden and the primary competing technical systematics are summarized below.

- **NDK transport-induced residual dephasing.** The predicted residual exponent is D_{pred}, with leading quadratic dependence on n, linear dependence on interrogation time T, and stronger dephasing for smaller pre-split coherence length $\lambda_{\mathrm{coh},0}$. Under the thermal identification, the same prediction becomes a linear dependence of the quadratic coefficient on source temperature Θ.
- **Ordinary pulse-infidelity accumulation.** This is expected to track pulse calibration and sequence fidelity more directly, and is typically linear in pulse number or otherwise implementation-specific rather than a clean residual n^2 law.
- **Wavefront and inhomogeneity effects.** These are generally tied to beam geometry, cloud expansion, spatial inhomogeneity, and apparatus-specific configuration. They may grow with LMT, but they are not expected to produce the same joint dependence on n^2, T, and pre-split coherence preparation.
- **Readout and detection noise.** These affect the uncertainty floor more directly than they generate an intrinsic residual coherence burden.

The useful discriminator is therefore not total contrast loss by itself, but a residual component that survives ordinary technical mitigation and retains the joint scaling

$$D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2 ,$$

or, under the thermal identification, a quadratic dephasing coefficient that scales linearly with Θ.

6 Falsifiers

The note is only useful if it states what would count against the proposal. At least three clean falsifiers are available.

(i) **No quadratic residual.** If contrast loss is fully captured by linear or calibration-driven models across the accessible LMT range, the transport-induced dephasing claim is disfavored.

(ii) **No preparation dependence.** If the residual loss does not track the pre-split coherence length $\lambda_{\mathrm{coh},0}$ or, under the thermal identification, the source temperature Θ in the predicted way, the closure-normalized transport prediction is disfavored.

(iii) **Disappearance under pulse optimization.** If the entire apparent signature is removed by standard pulse-engineering improvements, then the effect is better interpreted as ordinary technical loss.

A positive result would therefore not merely be "contrast falls at high LMT," but rather:

> a residual dephasing component that survives ordinary technical mitigation and scales jointly with n^2, T, and smaller pre-split coherence length $\lambda_{\mathrm{coh},0}$, or thermally with higher source temperature Θ.

7 Experimental value of the test

The experimental value of this proposal lies in its asymmetry. A positive result would point to a nonstandard coherence burden tied to matter-wave transport itself. A null result would still be highly informative by constraining the closure-normalized transport bridge and the specific route from the reduced transport theory to the laboratory forecast.

That is why MAGIS-100 is a particularly useful target. Even before ultimate design sensitivity is reached, the experiment can potentially probe the *shape* of the scaling law by controlled sweeps in LMT enhancement, preparation, and interrogation time.

8 Conclusion

The closure-normalized transport model predicts an additional dephasing exponent that can be stated in directly experimental variables:

$$D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2, \qquad \text{or thermally} \qquad D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2\, \Theta.$$

That prediction is sharper than the generic statement that large-momentum-transfer interferometers lose contrast at high pulse number. Its distinguishing content is the joint scaling with LMT enhancement, interrogation time, and smaller pre-split coherence length $\lambda_{\mathrm{coh},0}$, or thermally with higher source temperature Θ.

The action-scale closure establishes that the normalization of the predicted effect is fixed by the primitive-cell sector of the theory. Accordingly, the forecast presented here is parameter-free at the level of fundamental normalization, and the remaining uncertainty lies only in laboratory transfer factors rather than in the underlying action scale itself.

Accordingly, the most useful MAGIS-100 test is not a single headline measurement but a short discrimination program built from LMT sweeps, source-preparation sweeps, interrogation-time

sweeps, and ordinary pulse recalibration. The temperature sweep is especially valuable because it provides an additional derived control knob within the present bridge, rather than an auxiliary post hoc diagnostic. If a residual component with the proposed scaling survives those tests, it would provide direct evidence for the predicted matter-wave transport burden. If it does not, the closure-normalized transport bridge is correspondingly constrained.

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